

Exploring Trends in Food Spending

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Introduction

If you've found yourself "shell"shocked at grocery store egg prices in recent years, you're in good company. In fact, Americans express more concern regarding food prices than any other economic concern [10]. It's no wonder that food prices are top of mind: 2022 saw the highest inflation in food prices since the 1970s, with grocery prices rising 11.4% [1]. Through our interactive dashboard, we allow users to uncover trends in average U.S. food prices during the volatile period from 2019 through 2023. Our tool draws from extensive consumer grocery transaction data to visualize regional and time based trends across a range of common grocery items: eggs, chicken, ice cream, and coffee. Filtering capabilities within the dashboard will allow users to drill down into particular food items, years, and geographies. The share of wallet feature further illustrates consumer spending trends and explores relationships in spending across grocery retailers. Historic consumer data will be analyzed to 1) measure price elasticity via a regression model and 2) predict future price trends via a price forecasting model. Our analysis offers a comprehensive view into the forces shaping current food prices and allows users to make informed decisions amidst changing economic conditions.

Literature Survey

Food prices rose dramatically in the early 2020s, influenced by supply chain disruption, labor shortages, the COVID pandemic, and rising transportation costs [1, 4, 8, 14]. In particular, egg prices rose as a bird flu outbreak at the largest U.S. egg producer cut supply and drove up costs [3]. Although food prices are no longer growing at the pace they were during the pandemic, they remain high relative to pre-pandemic prices [5, 16]. Americans express significant concern over the price and affordability of food; 81% are worried about food prices, and 29% are worried about their ability to afford food for their household [2].

The USDA (U.S. Department of Agriculture) and ERS (Economic Research Service) monitor the key performance indicators such as individual, household, and market-level food consumption including prices, expenditures, food marketing costs, and farm-to-retail price differences [6, 13]. However, the data is typically derived from census samples and aggregated at higher levels, making granular data much harder to source and clean up. As a result, the aggregated data only allows us to observe price changes at a national level, failing to capture regional trends and consumer shopping patterns.

Similarly, policymakers rely on food price data to design and implement strategies that address price volatility. Detailed consumer shopping information can further assist government officials in crafting targeted policies that help everyday Americans shop more affordably [5, 9, 11, 16].

Additionally, the grocery merchants represented in our dataset would be interested in consumer trends that could directly influence the types of items they sell, their marketing strategies, and supply chain practices. Understanding the impact of prices on purchasing behavior may inform pricing strategies [17], and identifying stores which commonly share consumers may inform marketing approaches.

Data and Tools

Our primary dataset comes from Edison, a major e-receipt data provider that focuses on consumer panel data from grocery merchants, offering valuable insights into consumer loyalty and purchasing behavior. In addition to the Edison data source, we joined in data from the US Postal Service (on zip code) in order to get more geographical data such as state. We had a total of 5.8 million rows in

the raw dataset before any preprocessing was conducted. We narrowed the dataset down to 4 food categories: chicken, coffee pods & capsules, eggs, and frozen desserts. Data exploration and preprocessing was conducted in R, Python, and Snowflake, and the interactive dashboard was developed in Tableau and published to Tableau Public, accessible via a web browser.

Innovations & Methodology

The factors shaping a consumer's food environment --availability, affordability, convenience, and quality--play a crucial role in their purchasing decisions [12, 14]. Although our datasource is expansive, we are aware of the potential risk of a biased sample. Though existing research tracks broad trends in food prices, our dashboard is intended to offer a greater level of insight into consumer purchasing and the financial burden of grocery costs by drilling down into specific food items and consumer attributes. Retailers in our dataset will be particularly interested in these consumer-level trends, as they could influence product offerings, marketing strategies, and supply chain decisions. Additionally, these metrics reflect changes in overall buying power, helping retailers understand what their consumers can afford. Our food-focused Tableau dashboard consists of the following visual and analytical elements:

1. *Choropleth Map*: U.S. Map displaying average prices for selected food items in a given year.
2. *Share of Wallet*: Area plot displaying the relation between spending across different retailers.
3. *Price Forecaster*: Prediction of the selected food's future price, based on the ARIMA model.
4. *Price Elasticity*: Price elasticity curve with interactive filters for consumer income level.

We have investigated and built the models to inform our price elasticity curve, change detection, and price forecasting features. To capture price elasticity, we tested out different models using the following techniques: log-log transformation, Box-Cox transformation, and decision tree. Traditionally, economists use a log-log to estimate price elasticities by applying a logarithmic transformation to both price and quantity variables [17]. The benefits of this model include transforming the relationship between price and quantity into a linear one, with the price coefficient directly reflecting the elasticity of demand. This provides a straightforward and interpretable measure of how sensitive demand is to price changes [18]. In addition to the log-log model, we tested a model with Box-Cox transformation to stabilize variance and correct for skewed distributions since the response variable (quantity) was not normally distributed. We also experimented with a decision tree model to capture potential non-linear relationships in the data, allowing us to segment consumers based on factors like income, age, and gender, thus identifying how these factors influence the total amount spent in response to price changes.

Ultimately, our primary focus was on modeling price sensitivity based on income class. We had anticipated price sensitivity to vary across income classes due to differing impact of price changes: for instance, assuming higher-income consumers are likely less sensitive to price fluctuations, whereas lower-income consumers exhibit higher price sensitivity. Our regression model will capture the differential impact of income class by including coefficients, when added to the baseline price elasticity coefficient, will illustrate how each group's sensitivity to price changes differs.

Additionally, price forecasting has been explored using ARIMA, the common time series forecasting technique, with the intention of utilizing historic food prices to anticipate future trends in pricing. The ARIMA model can capture patterns in historical data including:

- Autoregressive components: incorporating past values to predict future ones
- Moving Average components: Using past forecast errors to refine predictions, thereby addressing volatility in price changes.

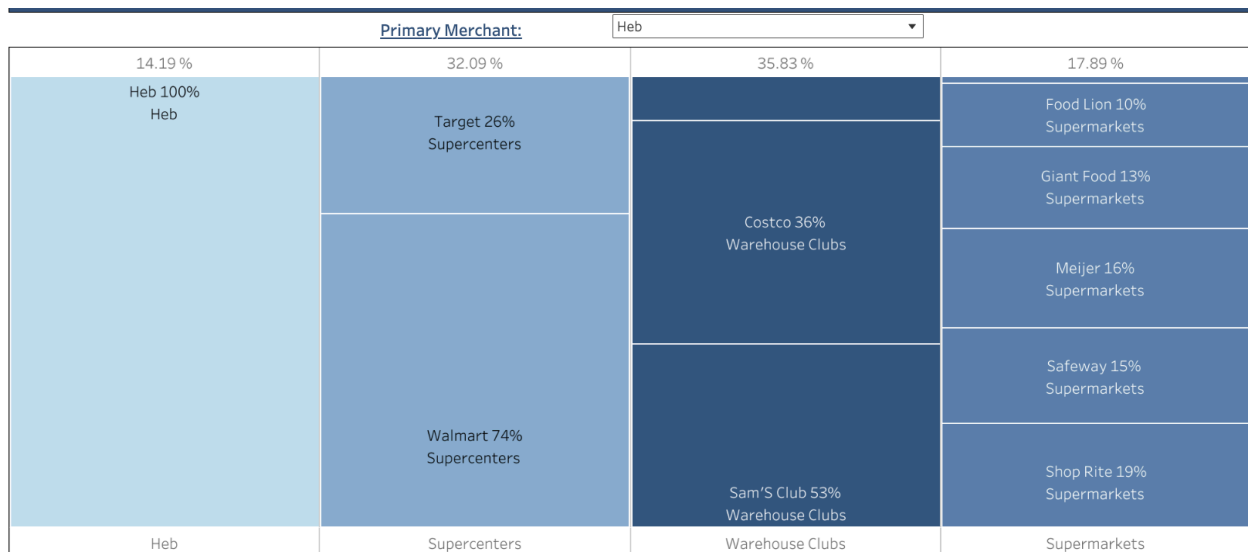
The flexibility of ARIMA makes it ideal for analyzing food price fluctuations, which often exhibit short-term irregularities, due to seasonal effects or supply chain disruptions, while maintaining long term

trends [4]. The ARIMA model was trained on historical monthly averages for the selected food items, and using the fitted model, we projected future price trends over a 24 month period.

Experiments & Evaluation

We conducted initial experiments to evaluate the accuracy of our time series forecasting model for predicting future trends and to assess the robustness of our regression equation in estimating price elasticity under varying market conditions. In addition, the team created a share of wallet analysis and a choropleth map to explore food price trends across the United States. The choropleth map helps us analyze price trends, while the price elasticity and share of wallet analyses help us understand consumer behavior. We aimed to answer this set of questions:

- How much of a difference is there in price elasticity for higher range income classes?
- Can we use our historical food price data to predict future prices?
- In the grocery category, which merchants dominate the consumer's share of wallet?
- To what degree does the price of staple food items vary from state to state and region to region?



The share of wallet analysis was crucial in our study of consumer behavior, focusing on the proportion of consumer spending distributed among various grocery merchants. The analysis enables users to identify which merchants dominate consumer spending, and if the share trends are consistent as you change the primary merchant of reference. What we observed is that share among the categories of Supercenters, Warehouse Clubs, and Supermarkets stay relatively consistent between 30-35% share, but there are some merchants that dominate the overall share, irrespective of category such as Walmart, Sam's Club, HEB, and Costco, with 24%, 19%, 14%, and 13%, respectively, of the consumer's overall share.

In our baseline elasticity model, we aimed to predict item quantity based on item price and income class using a log-log model, which is commonly used for estimating price elasticity. After applying log transformations to item quantity and item price, the model yielded a low R-squared value of 0.05027. Further investigation revealed that the item quantity field was heavily skewed towards a value of 1, with minimal variability, which likely contributed to the poor model performance. To address the lack of variability in item quantity, we shifted our focus to order total amount to examine whether changes in item prices could influence overall spending per order, thereby allowing us to measure price elasticity at

the level of total purchase amounts. Although the order total amount had more variability, it was right-skewed, leading us to apply a Box-Cox transformation with an optimal lambda of 0.5 (square root transformation). We included additional predictors such as age, item category, and gender to the second model to see if more variability would be explained. The model's R-squared remained low at 0.01812, suggesting that linear models might not capture the underlying complexity.

As a next step, we pivoted to a decision tree model to capture non-linear patterns. Since R(software) does not provide R-squared in its standard output, we used mean absolute error (MAE) as the accuracy measure for the decision tree. We used the same variables that were used in the Box-Cox transformed model. While the model was able to split on sales per item, income class, and age group variables, the MAE was 69.13 units. With the median order total amount being 122.77, this indicates low accuracy for this model. Given these challenges, we introduced a new variable: average price per unit, order amount total divided by order quantity, to be the response variable in our subsequent model. Average price per unit has a moderately negative correlation (-0.37) with order total quantity. We then fit a log-log model using `log_quantity` as the response variable and `log_avg_price` as the predictor,

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--- Price Elasticity Model Summary ---

Call:
lm(formula = log_quantity ~ log_avg_price + USER_INCOME_CLASS,
    data = df)

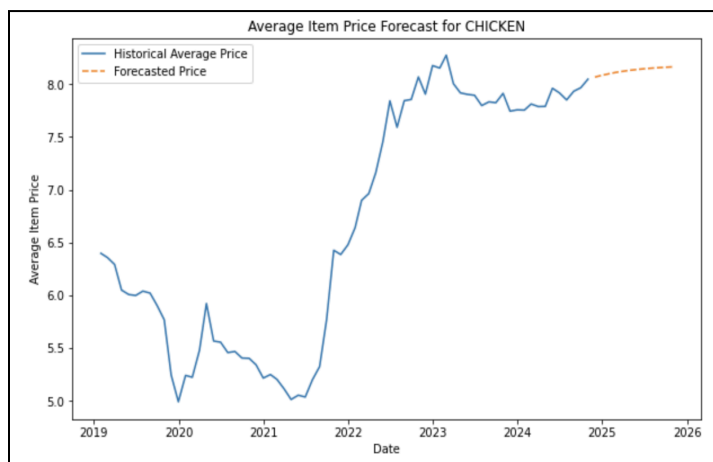
Residuals:
    Min       1Q   Median       3Q      Max
-4.0426 -0.4093  0.0421  0.4508  4.6892

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  4.796816   0.004154 1154.773 <2e-16 ***
log_avg_price -0.881912   0.001499 -588.377 <2e-16 ***
USER_INCOME_CLASS0-25k -0.053508   0.004318  -12.392 <2e-16 ***
USER_INCOME_CLASS100k-150k  0.037032   0.003569   10.375 <2e-16 ***
USER_INCOME_CLASS150k-200k  0.040758   0.004044   10.079 <2e-16 ***
USER_INCOME_CLASS200k+  0.074038   0.003855   19.206 <2e-16 ***
USER_INCOME_CLASS25k-50k -0.039820   0.003836  -10.380 <2e-16 ***
USER_INCOME_CLASS50k-75k -0.002256   0.003685   -0.612  0.5404
USER_INCOME_CLASS75k-100k  0.006785   0.003692   1.838  0.0661 .
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Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6336 on 659599 degrees of freedom
Multiple R-squared:  0.3449, Adjusted R-squared:  0.3449
F-statistic: 4.342e+04 on 8 and 659599 DF, p-value: < 2.2e-16

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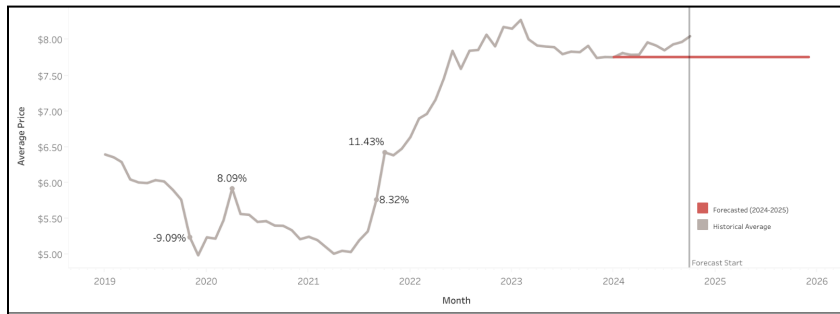
along with user income class. This model achieved an improved R-squared of 0.3449. The coefficient for `log_avg_price` was -0.879, indicating inelastic price elasticity—a 1% increase in average price corresponds to an approximate 0.88% decrease in quantity purchased. The model also accounts for the impact of income class on price elasticity. Coefficient estimates show that lower-income classes decrease the `log_avg_price` coefficient, making them more sensitive to price changes, while higher-income classes increase the coefficient, reflecting less sensitivity to price fluctuations. The F-statistic of 43,420 confirmed the model's statistical significance, indicating that the selected predictors effectively explain the variation in quantity purchased.



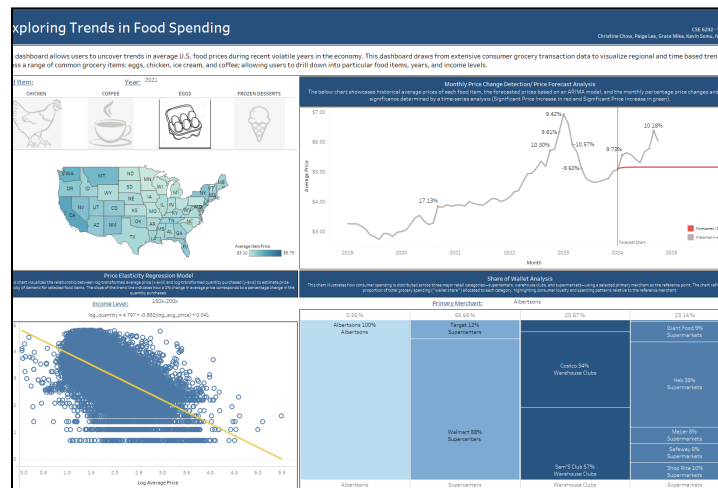
In addition to the price elasticity model, our team also built a time series model. In the example shown to the left, the ARIMA model forecasts the average monthly price of chicken over a 12-month period. We first re-sampled the data to obtain monthly averages and then selected an $ARIMA(1,1,1)$ to fit the historical data and create a forecast. The forecast supports our literature review findings: food prices continue to rise, but at a slower pace than

the later stages of the pandemic when they rose steeply.

In addition to creating ARIMA models for all the other food categories, we changed the way we trained and fit the model. In the first iteration of ARIMA modeling, we used all the historical data up until mid 2024. In our new iteration, we decided to train the model only up until the end of 2023 and compare our 2024 forecasts with our historical 2024 data. The team also spent time fine tuning the hyperparameters using the pmdarima package and ended up choosing a unique and optimal p,d,q for each



food category. The selected ARIMA(0,1,0) model, chosen based on AIC minimization, forecasts a relatively steady trend for 2024-2025. The team also calculated month-over-month percent changes in historical price and highlighted specific months in which there were significant increases or decreases.



Informed by early data exploration, our team had initially proposed eggs and french toast as our focal food items, as these purchases showed up with the highest frequency in our original food delivery dataset. However, more in depth data exploration revealed issues with our dataset and prompted a pivot to a grocery dataset that would better fit our project's goal. We ended up choosing a total of four categories to focus on: chicken, coffee pods & capsules, eggs, and frozen desserts, in order to explore whether price trends were consistent across common basic food

items and more "luxury" food indulgences. To further enable our understanding of how these data points interact with each other, we built a Tableau dashboard based on our analyses above. We incorporated interactivity within the dashboard by allowing the user to select from among the aforementioned food types, as well as toggle filters for purchase year and the consumer demographic of income level. Upon selecting these filters, we have generated the following visuals and features: 1) a price forecaster to predict the next year's price of the given food item with significant spikes in historical price highlighted, 2) an area chart showcasing the share of wallet, 3), a price elasticity analysis and 4) a choropleth map of the United States displaying a gradient of the average prices of the food categories selected. The interactivity of these different charts allows for users to visually see how specific indicators such as year, food item and income level affect the spending power of individuals over time.

Upon initial completion of our dashboard, we wanted to assess the usability and effectiveness of the dashboard. We conducted user experience testing by surveying 12 respondents via a survey which

included objective questions prompting users to engage specific filters and interpret results, as well as subjective questions gauging the ease of use of various dashboard components. The majority of participants correctly answered questions pertaining to their interpretation of the dashboard, although some experienced difficulty in understanding various dashboard components. Participants provided feedback on suggested improvements, allowing us to gain insights into both the interpretability of the visualizations and the overall user experience. This feedback was used to guide refinements to enhance clarity, functionality, and accessibility of the dashboard's tools. Refinements based on feedback included making certain filters more centralized as opposed to having spread throughout the dashboard, updating the dashboard to have more of a neutral color scheme that allowed for easily spotting outliers or changes across states or time periods, and using dynamic scaling of our legends as opposed to making our legends static for all four food items (specifically for the choropleth map). Our intentions for the final dashboard are to offer a compilation of tools through which consumers and retailers alike can draw valuable insights about the American food environment. A deeper understanding of food price trends and consumer spending patterns can inform personal grocery budgets or grocery retail business strategy.

Conclusion

Upon completion of this effort with the intention of building out a tool to be used by retailers and grocery suppliers, the team uncovered a number of insights into the complexities of consumer spending on food items. For example, upon examining the choropleth map, it's noticeable that prices tend to skew higher closer to states along the coasts, Alaska, and Hawaii. This is reasonable due to much of the agricultural production taking place in the midwestern region of the US. And that higher prices tend to align with states known for their higher cost of living, as well as for states that need food to be imported.

The final price elasticity model was able to explain 34.39% of the variation of the logarithm of order total quantity. Although this R-squared value indicates not all variability in quantity is explained by the model, it is reasonable given the complexity of consumer behavior, and the fact that not all external factors were incorporated into the model. Marketing promotions, economic conditions, and seasonal trends were not explicitly incorporated in the model, and addressing these could improve the R-squared value. The price elasticity coefficient is -0.88 , which translates to inelastic demand for grocery food items. In addition, lower income classes (e.g., "0-25K" and "25K-50K") have negative coefficients indicating greater price sensitivity compared to the baseline, while higher income classes (e.g., "100K-150K", "150K-200K", "200K+") have positive coefficients, suggesting less sensitivity to price changes.

The share of wallet analysis confirmed our initial hypotheses that large supermarkets and warehouse clubs would dominate a consumer's share of spend, including Walmart, Sam's Club, and Costco. Further analysis could be done here to look at the share of wallet at different aggregations of food categories and specific income levels to reveal trends at these levels.

While the ARIMA model was a good first approach to time series forecasting, its performance highlights some limitations. The model's simplicity might have oversimplified the complex dynamics of food prices, as it does not account for seasonality, external variables, or potential structural changes in the data. The visualization also shows that ARIMA predictions do not align perfectly with historical patterns, suggesting that another type of model might yield better results. In future iterations, we would like to explore the seasonal ARIMA model, exponential smoothing, and XGBoost models and compare results.

All team members contributed evenly to this project, ensuring a balanced and collaborative effort. We hope our tool is a robust, data-driven solution that can provide meaningful value to grocery retailers, the public, and policy makers in understanding and navigating pricing and consumer behavior.

Appendix

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